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Temporal Segregation in Coexisting *Acomys* Species: The Role of Odour

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HAIM, A. AND F. M. ROZENFELD. *Temporal segregation in coexisting Acomys species: The role of odour.* PHYSIOL BEHAV 54(6) 1159–1161, 1993.—To understand the mechanisms underlying displacement (the shift from nocturnal to diurnal activity), in one of the two coexisting spiny mice (genus *Acomys*), the effect of chemical cues released by *A. cahirinus* on the time of activity of *A. russatus* was tested Six golden spiny mice (*A. russatus*), which prior to the experiments were kept separate from common spiny mice (*A. cahirinus*), showed nocturnal activity. They were exposed to chemical cues from the urine and faeces of conspecific and heterospecific mice of the opposite sex. The onset of activity in these mice was recorded. While the urine and faeces of conspecific mice did not have a significant effect on the time of onset of activity, heterospecific urine and faeces did cause a significant ($p < 0.001$) time shift and, a day after they were introduced, activity started 6.8 ± 1.9 h earlier. This shift also took place on the second day. The results of this study suggest that the mechanism for displacement of *A. russatus* from nocturnal activity is by chemical signals released by *A. cahirinus*. Therefore, it may be concluded that chemical cues maintain time separation between these two species.

Coexisting Chemical signals Activity Odour Arid Rodent Temporal segregation

THE spiny mice of the genus *Acomys* are represented in Israel by two species: the common spiny mouse *A. cahirinus* and the golden spiny mouse *A. russatus*. Both species are rock dwellers. While *A. cahirinus* is widely distributed throughout the country ranging from the Mediterranean woodlands up to the extreme hot and arid habitats of the Israeli Rift Valley and the Negev desert, *A. russatus* is limited in its distribution to the extremely arid regions. Consequently, in the most extreme hot and arid parts of the country these two species are sympatric (4,12).

The coexistence of these two species is possible due to displacement of *A. russatus* from nocturnal activity into diurnal activity by *A. cahirinus* (12). Shkolnik (13) demonstrated experimentally that removal of *A. cahirinus* from the common habitat resulted in a change of activity pattern of *A. russatus*, which became nocturnal. Recently, it has been demonstrated that when *A. russatus* is kept separately from *A. cahirinus* under laboratory conditions it exhibits the activity pattern of a nocturnal rodent (10). Most small mammals, like rodents, inhabiting hot and arid environments avoid exposure to excessive heat by nocturnal activity (11). The adaptations to diurnal habits of *A. russatus* in an extremely hot and arid environment are, therefore, of great interest (14).

Although the phenomenon of displacement is well documented in the sympatric *Acomys* species, its behavioural mechanisms, as well as the resource for which these two species are competing, remain to be explained. Recently, it was shown that odour cues released by *A. cahirinus* altered the daily rhythms of oxygen consumption of *A. russatus* (3). This suggests that activity patterns of *A. russatus* may be affected by olfactory signals. It may be hypothesized that chemical signals of *A. cahirinus* released into the common environment play a role in the displacement of *A. russatus* from nocturnal activity.

The aim of this study was to test the above hypotheses under laboratory conditions. Therefore, we investigated the effect of urine and faeces released by *A. cahirinus* on the activity patterns of nocturnal *A. russatus* kept apart from *A. cahirinus*. The results of such a study may help to assess the mechanism for displacement of *A. russatus* from nocturnal activity by *A. cahirinus*.

METHOD

Six *A. russatus* (four males and two females, mean body mass 63 g) and three *A. cahirinus* (one male and two females, mean body mass 37 g) were used. The animals of both species were obtained from the breeding colonies (The University of Haifa at Oranim) established by animals brought from the Dead Sea area. The mice were transported to Belgium in separate cages to avoid chemical communication.

Prior to the laboratory trials, animals of the two species were kept in separate rooms, each mouse in a polycarbonate cage (28 × 26 × 16 cm) at an ambient temperature of 27°C and under a photoperiod regime of 12L:12D. Neon lights were on between 0700 and 1900 h, and during the dark phase a constant dim orange light was maintained, so that continuous observations could be carried out. *A. russatus* were used as experimental subjects as well as stimulus producers, while *A. cahirinus* were used

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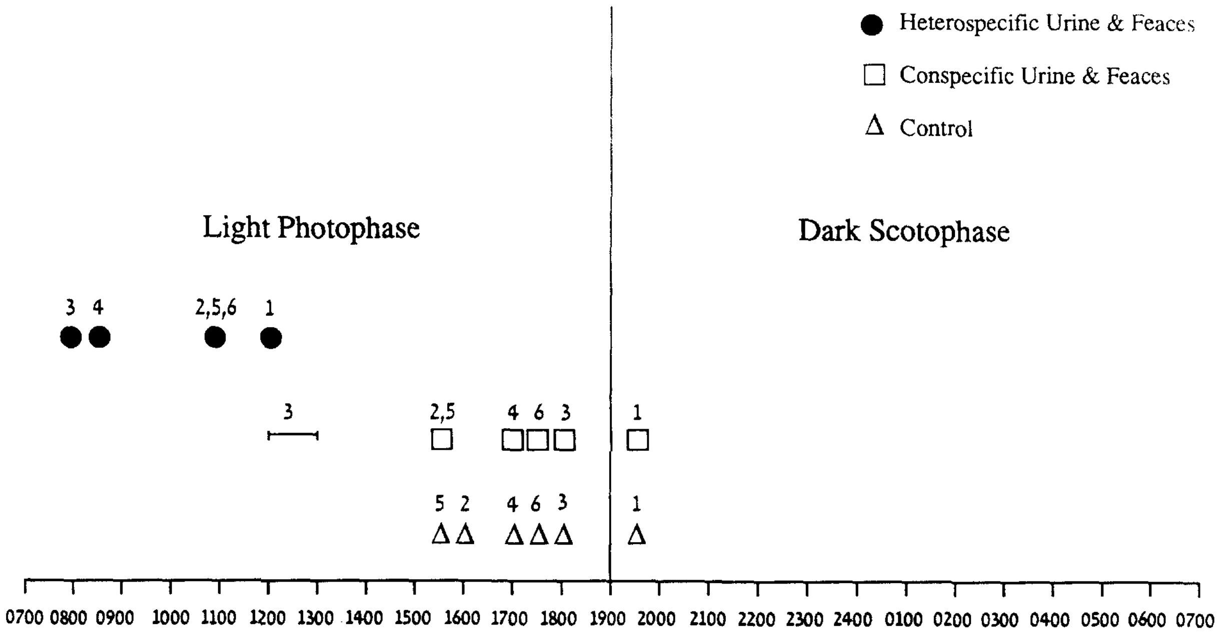

FIG. 1. Onset of activity in golden spiny mice *Acomys russatus*. The symbols indicate the hour in which activity started in each animal. The number of each animal is marked above the symbols. The time shift, in hours, due to exposure of *A. russatus* to heterospecific urine and feaces, is highly significant. A short burst of activity of female No. 3 at midday, indicated by a horizontal line, was anomalous (see text).

only as stimulus producers. The mice were kept on a diet of sunflower seeds and commercial hamster chow, and carrots and cucumbers were used as the only source of water. During experiments they were kept on a diet of only sunflower seeds and carrots.

Each tested *A. russatus* was housed in a modular aluminium observation pen (57 × 57 × 48 cm) connected to a Y-shaped nesting place made of plaster and covered with glass (8,9). The pen had one glass wall for observation and was covered with a wire mesh. The paired branches of each nesting place were connected to the observation pen, while the stem was covered by a wire mesh. In such an arrangement, the studied mouse could use two entrances to its nesting place. The floors of both observation pen and nesting place were lined with filter paper (Schleicher and Schull) (9).

Wire mesh cages were used to collect urine and faeces. The cages were placed above a sheet of parafilm. Urine and faeces were removed from the parafilm into tubes by using a clean pipette glass. Samples from each mouse were kept separately in a sealed tube at 3°C.

Experimental Procedure

Tests were conducted in the room in which *A. russatus* were kept. Each test lasted 4 consecutive days. For the first day, two animals were put simultaneously into separate observation pens. The mice were put into the nesting places that were connected to the observation pens at 1100 h. Sunflower seeds and carrots were offered in a glass bowl inside the pen. After 30 min of habituation, the mice were observed continuously until the onset of activity. The time that activity started was recorded and the behaviour of each mouse was videotaped. The results of day 1 were used as the data for control.

On the second day at 1100 h, the floor filter paper was changed and fresh food was added. For control, 12 drops of urine and 16 faecal pellets of the conspecific opposite sex were deposited at the center of each observation pen. Recording was as in the first day.

On the third day at 1100 h, the filter paper was changed and food was added. Twelve drops of urine and 16 faeces pellets of *A. cahirinus* of the opposite sex were added at the centre of the observation pen. Recording continued for a further 48 h, during which the filter paper was not changed and only fresh carrots were added.

Experiments were conducted in August and September 1990. Results are given as mean ± SD. The differences in onset of activity were tested by using the paired t-test.

RESULTS

The onset of activity in five out of the six naive *A. russatus* was within 3.5 h prior to the lights being turned off, while one animal (No. 1, Fig. 1) began its activity 30 min after darkness. All animals were active in the dark phase, activity stopping after midnight and resuming on the afternoon of the next day. Such a pattern of activity was also observed in other animals in their cages kept in the same room. The exposure to heterosexual conspecific urine and faeces did not induce a significant change ($t = 1.58$, $p > 0.1$, NS) in onset of activity. Animal No. 2 began its activity 30 min earlier than on the previous day. Animal No. 3 entered the observation pen for about 20 min between 1200 and 1300 h to sniff the conspecific urine and faeces. However, at 1800 h it began its activity, which continued until midnight.

Adding heterospecific urine and faeces to the tested animals resulted in a highly significant shift ($t = 9.12$, $p < 0.001$) in the time of onset of activity. All animals began their activity earlier on the day following the introduction of the stimulus. The average rate of the shift was 6.8 ± 1.9 h compared with the second day, in which the mice were exposed to conspecific urine and faeces (Fig. 1). This temporal shift in activity was repeated on the following day.

DISCUSSION

The competitive exclusion principle formulated by Gause (2) makes it clear that two species of the same genus cannot occupy the same niche. Furthermore, the displacement is such that one of the species can take possession of a particular mode of life or kind of food, so that it will have an advantage over its competitor. The coexistence of the two *Acomys* species in the same habitat may be a clear example of Gause's principle.

The two *Acomys* species are sympatric in the most hot and arid parts of the Israeli Rift Valley and in the Negev desert (4,12). The desert environment is characterized by low rainfall, which is highly variable and unpredictable. Water is the most important controlling factor for many biological processes occurring in this ecosystem (7). To cope with such a hostile habitat, small mammals, such as rodents, are nocturnal and select suitable microhabitats for the daytime, such as burrows or under rocks. Such a behavioural pattern allows them to escape the daytime heat and to conserve water. One of the best known examples is the Kangaroo rat, *Dipodomys merriami*, which is common in the deserts of North America (11). The diurnal activity of the golden spiny mouse seems to be an exception. Its activity pattern results from displacement from nocturnal activity by *A. cahirinus* (13). Diurnal patterns of activity are reported for other Muridae species such as *Lemniscomys griselda* and *Rhabdomys pomillio* (15). While the former is limited in its distribution to mesic habitats in the Southern African subregion, the latter also inhabits extremely hot and dry deserts like the Namib. Both species, unlike *A. russatus*, are burrow dwellers (15).

The physiological adaptations of *A. russatus*, which enable it to cope with its hostile environment, even as a diurnal rodent, were demonstrated in a comparative study of these two species (12,14). These include high thermoregulatory capacity at high ambient temperatures, resting metabolic rates 44% lower than predicted for their body mass according to Kleiber's equation (5), and a higher concentrating capacity of the kidney compared with *A. cahirinus*. The dark pigmentation on the skin of *A. russatus* also seems to be an important adaptation, as it reduces penetration of destructive radiation.

The present study shows that *A. russatus*, when isolated from *A. cahirinus*, become nocturnal, as already observed in field studies (13) and under laboratory conditions (10). The isolated *A. russatus* started being active in the afternoon and throughout the night (Fig. 1). Odour cues from unknown conspecific mice of both sexes did not affect the onset of activity in nocturnal, isolated *A. russatus*. In the golden hamster, *Mesocricetus auratus*, under free-running rhythm conditions, conspecific odour cue interactions did cause a phase shift in its activity (6). The introduction of heterospecific urine and faeces into the environment of *A. russatus* caused a significant ($p < 0.001$) shift of 6.8 ± 1.9 h in the time in which activity started, and such mice displayed a pattern of relative diurnal activity. The addition of these same chemical signals released by *A. cahirinus* into a metabolic chamber in which *A. russatus* was kept separately at a photoperiod of 12L:12D caused a shift in the rhythm of oxygen consumption (3) as well as in the rhythm of body temperature (1).

The activity of female No. 3 for a short period (20 min) at midday could probably be related to her reproductive status. At the end of the experiment, the two tested females were paired with a male and only female No. 3 mated immediately. Therefore, it is possible that she reacted in such a way to the odours of a conspecific male because she was in a receptive state. However, so far nothing is known about the reproductive physiology of this species.

The results of the present study may explain the observations of Shkolnik (13), in which the removal of *A. cahirinus* from the common habitat in nature resulted in nocturnal activity of *A. russatus*, and suggest that chemical cues released by *A. cahirinus* to the common habitat may maintain the time separation of the activity patterns of these two species. They also demonstrate that, in rodents, chemical signals may play an important role in the mechanisms that temporally segregate two sympatric species. The chemical source of such cues remains unknown. Further investigation is needed to determine if any, or only special, new odours elicit this type of response and if habituation to such odours occurs. However, in mammals, urine and/or faeces are common vehicles for olfactory communication and so it would be reasonable to infer that they play a role in determining the observed behaviour.

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